

Assignment 2 (ML for TS) - MVA

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1 Introduction

Objective. The goal is to better understand the properties of AR and MA processes and do signal denoising with sparse coding.

Warning and advice.

- Use code from the tutorials as well as from other sources. Do not code yourself well-known procedures (e.g., cross-validation or k-means); use an existing implementation.
- The associated notebook contains some hints and several helper functions.
- Be concise. Answers are not expected to be longer than a few sentences (omitting calculations).

Instructions.

- Fill in your names and emails at the top of the document.
- Hand in your report (one per pair of students) by Sunday 7th December 11:59 PM.
- Rename your report and notebook as follows:
FirstnameLastname1_FirstnameLastname1.pdf and
FirstnameLastname2_FirstnameLastname2.ipynb.
For instance, LaurentOudre_ValerioGuerrini.pdf.
- Upload your report (PDF file) and notebook (IPYNB file) using this link:
<https://forms.gle/J1pdeHspSs9zNfWAA>.

2 General questions

A time series $\{y_t\}_t$ is a single realisation of a random process $\{Y_t\}_t$ defined on the probability space $(\Omega, \mathcal{F}, P)$, i.e. $y_t = Y_t(w)$ for a given $w \in \Omega$. In classical statistics, several independent realizations are often needed to obtain a “good” estimate (meaning consistent) of the parameters of the process. However, thanks to a stationarity hypothesis and a “short-memory” hypothesis, it is still possible to make “good” estimates. The following question illustrates this fact.

Question 1

An estimator $\hat{\theta}_n$ is consistent if it converges in probability when the number n of samples grows to ∞ to the true value $\theta \in \mathbb{R}$ of a parameter, i.e. $\hat{\theta}_n \xrightarrow{\mathcal{D}} \theta$.

- Recall the rate of convergence of the sample mean for i.i.d. random variables with finite variance.
- Let $\{Y_t\}_{t \geq 1}$ a wide-sense stationary process such that $\sum_k |\gamma(k)| < +\infty$. Show that the sample mean $\bar{Y}_n = (Y_1 + \dots + Y_n)/n$ is consistent and enjoys the same rate of convergence as the i.i.d. case. (Hint: bound $\mathbb{E}[(\bar{Y}_n - \mu)^2]$ with the $\gamma(k)$ and recall that convergence in L_2 implies convergence in probability.)

Answer 1

1.1.

The sample mean is defined as

$$\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i.$$

Consistency (Law of Large Numbers):

$$\bar{Y}_n \xrightarrow{P} \mu \quad \text{as } n \rightarrow \infty.$$

Rate of Convergence (Central Limit Theorem):

$$\sqrt{n}(\bar{Y}_n - \mu) \xrightarrow{\mathcal{D}} \mathcal{N}(0, \sigma^2),$$

which implies

$$\bar{Y}_n - \mu = O_p\left(\frac{1}{\sqrt{n}}\right).$$

1.2.

We first compute the variance of the sample mean

Since the process is WSS, we have

$$\text{Var}(\bar{Y}_n) = \text{Var}\left(\frac{1}{n} \sum_{t=1}^n Y_t\right) = \frac{1}{n^2} \sum_{t=1}^n \sum_{s=1}^n \text{Cov}(Y_t, Y_s) = \frac{1}{n^2} \sum_{t=1}^n \sum_{s=1}^n \gamma(t-s).$$

Let's now try to bound the variance using the absolute sum of autocovariances

We can rewrite

$$\text{Var}(\bar{Y}_n) = \frac{1}{n^2} \sum_{k=-(n-1)}^{n-1} (n - |k|) \gamma(k)$$

where $k = t - s$ and each lag k appears $n - |k|$ times. Taking absolute values:

$$\text{Var}(\bar{Y}_n) \leq \frac{1}{n^2} \sum_{k=-(n-1)}^{n-1} n |\gamma(k)| \leq \frac{1}{n} \sum_{k=-\infty}^{\infty} |\gamma(k)|.$$

Since $\sum_{k=-\infty}^{\infty} |\gamma(k)| < \infty$, there exists a constant $C > 0$ such that

$$\text{Var}(\bar{Y}_n) \leq \frac{C}{n}.$$

And,

$$\mathbb{E}[(\bar{Y}_n - \mu)^2] = \text{Var}(\bar{Y}_n) \leq \frac{C}{n} \longrightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since convergence in L^2 implies convergence in probability, we conclude

$$\bar{Y}_n \xrightarrow{P} \mu.$$

Finally, from the bound on the variance,

$$\text{Var}(\bar{Y}_n) = O\left(\frac{1}{n}\right),$$

which implies

$$\bar{Y}_n - \mu = O_p\left(\frac{1}{\sqrt{n}}\right),$$

so the sample mean enjoys the **same rate of convergence** as in the i.i.d. case.

3 AR and MA processes

Question 2 Infinite order moving average MA(∞)

Let $\{Y_t\}_{t \geq 0}$ be a random process defined by

$$Y_t = \varepsilon_t + \psi_1 \varepsilon_{t-1} + \psi_2 \varepsilon_{t-2} + \cdots = \sum_{k=0}^{\infty} \psi_k \varepsilon_{t-k} \quad (1)$$

where $(\psi_k)_{k \geq 0} \subset \mathbb{R}$ ($\psi = 1$) are square summable, i.e. $\sum_k \psi_k^2 < \infty$ and $\{\varepsilon_t\}_t$ is a zero mean white noise of variance σ_ε^2 . (Here, the infinite sum of random variables is the limit in L_2 of the partial sums.)

- Derive $\mathbb{E}(Y_t)$ and $\mathbb{E}(Y_t Y_{t-k})$. Is this process weakly stationary?
- Show that the power spectrum of $\{Y_t\}_t$ is $S(f) = \sigma_\varepsilon^2 |\phi(e^{-2\pi i f})|^2$ where $\phi(z) = \sum_j \psi_j z^j$. (Assume a sampling frequency of 1 Hz.)

The process $\{Y_t\}_t$ is a moving average of infinite order. Wold's theorem states that any weakly stationary process can be written as the sum of the deterministic process and a stochastic process which has the form (1).

Answer 2

2.1

Since $\mathbb{E}[\varepsilon_t] = 0$ and linearity of expectation, we have

$$\mathbb{E}[Y_t] = \mathbb{E}\left[\sum_{k=0}^{\infty} \psi_k \varepsilon_{t-k}\right] = \sum_{k=0}^{\infty} \psi_k \underbrace{\mathbb{E}[\varepsilon_{t-k}]}_0 = 0.$$

$$\boxed{\mathbb{E}[Y_t] = 0}$$

Autocovariance of Y_t

For $h \geq 0$, consider

$$\gamma_Y(h) = \text{Cov}(Y_t, Y_{t-h}) = \mathbb{E}[Y_t Y_{t-h}] - \mathbb{E}[Y_t] \mathbb{E}[Y_{t-h}] = \mathbb{E}[Y_t Y_{t-h}].$$

Compute the product:

$$Y_t Y_{t-h} = \left(\sum_{k=0}^{\infty} \psi_k \varepsilon_{t-k}\right) \left(\sum_{j=0}^{\infty} \psi_j \varepsilon_{t-h-j}\right) = \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \psi_k \psi_j \varepsilon_{t-k} \varepsilon_{t-h-j}.$$

Taking expectation, using that ε_t is white noise ($\mathbb{E}[\varepsilon_t \varepsilon_s] = 0$ for $t \neq s$, σ_ε^2 for $t = s$):

$$\mathbb{E}[Y_t Y_{t-h}] = \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \psi_k \psi_j \underbrace{\mathbb{E}[\varepsilon_{t-k} \varepsilon_{t-h-j}]}_{\sigma_\varepsilon^2 \mathbf{1}_{\{k-j=h\}}} = \sigma_\varepsilon^2 \sum_{k=0}^{\infty} \psi_k \psi_{k-h},$$

where $\psi_m = 0$ if $m < 0$. Therefore, the autocovariance function is:

$$\gamma_Y(h) = \sigma_\varepsilon^2 \sum_{k=0}^{\infty} \psi_k \psi_{k-h}, \quad h \geq 0, \quad \gamma_Y(-h) = \gamma_Y(h)$$

—

Weak stationarity

A process is weakly stationary if:

1. $\mathbb{E}[Y_t]$ is constant in $t \rightarrow \text{true}$, $\mathbb{E}[Y_t] = 0$. 2. $\text{Cov}(Y_t, Y_{t-h}) = \gamma_Y(h)$ depends only on lag $h \rightarrow \text{true}$, as shown above. 3. $\text{Var}(Y_t) = \gamma_Y(0) < \infty \rightarrow \text{true}$, because $\sum_k \psi_k^2 < \infty$ and $\text{Var}(Y_t) = \sigma_\varepsilon^2 \sum_k \psi_k^2 < \infty$.

$\implies \{Y_t\}$ is a **weakly stationary** process.

2.2

Power Spectrum of MA(∞) process:

The autocovariance of the MA(∞) process is

$$\gamma_Y(h) = \sigma_\varepsilon^2 \sum_{k=0}^{\infty} \psi_k \psi_{k-|h|}.$$

The power spectrum is the Fourier transform:

$$S(f) = \sum_{h=-\infty}^{\infty} \gamma_Y(h) e^{-2\pi i f h} = \sigma_\varepsilon^2 \sum_{k=0}^{\infty} \psi_k \sum_{j=0}^{\infty} \psi_j e^{-2\pi i f (k-j)}.$$

Factor the sums:

$$S(f) = \sigma_\varepsilon^2 \left(\sum_{k=0}^{\infty} \psi_k e^{-2\pi i f k} \right) \left(\sum_{j=0}^{\infty} \psi_j e^{-2\pi i f j} \right)^* = \sigma_\varepsilon^2 |\phi(e^{-2\pi i f})|^2,$$

Question 3 AR(2) process

Let $\{Y_t\}_{t \geq 1}$ be an AR(2) process, i.e.

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t \quad (2)$$

with $\phi_1, \phi_2 \in \mathbb{R}$. The associated characteristic polynomial is $\phi(z) := 1 - \phi_1 z - \phi_2 z^2$. Assume that ϕ has two distinct roots (possibly complex) r_1 and r_2 such that $|r_i| > 1$. Properties on the roots of this polynomial drive the behavior of this process.

- Express the autocovariance coefficients $\gamma(\tau)$ using the roots r_1 and r_2 .
- Figure 1 shows the correlograms of two different AR(2) processes. Can you tell which one has complex roots and which one has real roots?
- Express the power spectrum $S(f)$ (assume the sampling frequency is 1 Hz) using $\phi(\cdot)$.
- Choose ϕ_1 and ϕ_2 such that the characteristic polynomial has two complex conjugate roots of norm $r = 1.05$ and phase $\theta = 2\pi/6$. Simulate the process $\{Y_t\}_t$ (with $n = 2000$) and display the signal and the periodogram (use a smooth estimator) on Figure 2. What do you observe?

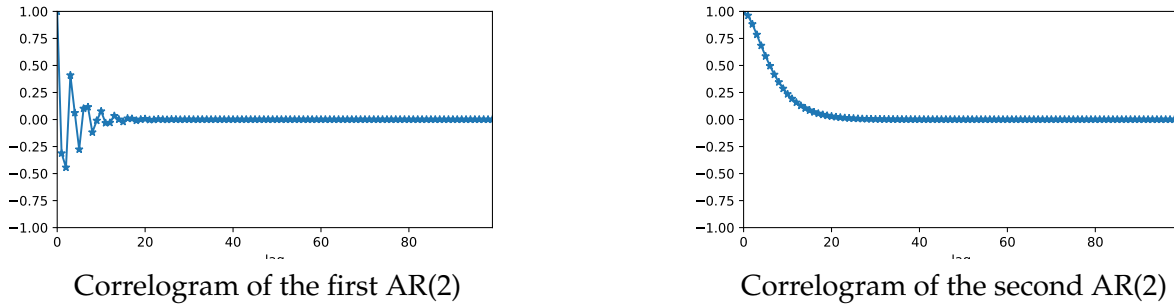


Figure 1: Two AR(2) processes

Answer 3

3.1

The autocovariance function $\gamma(\tau) = \text{Cov}(Y_t, Y_{t-\tau})$ satisfies the Yule-Walker recursion:

$$\gamma(\tau) = \phi_1 \gamma(\tau - 1) + \phi_2 \gamma(\tau - 2), \quad \tau \geq 1,$$

with $\gamma(0)$ determined by

$$\gamma(0) = \phi_1 \gamma(1) + \phi_2 \gamma(2) + \sigma_\varepsilon^2.$$

The general solution of the second-order linear difference equation is:

$$\gamma(\tau) = A r_1^{-|\tau|} + B r_2^{-|\tau|}, \quad \tau \in \mathbb{Z},$$

where A and B are constants determined by $\gamma(0)$ and $\gamma(1)$.

Using $\gamma(0)$ and $\gamma(1)$:

$$\begin{cases} \gamma(0) = A + B, \\ \gamma(1) = Ar_1^{-1} + Br_2^{-1}, \end{cases} \Rightarrow A = \frac{\gamma(1) - \gamma(0)/r_2}{1/r_1 - 1/r_2}, \quad B = \gamma(0) - A.$$

Thus the autocovariance coefficients are completely determined by the roots r_1, r_2 and σ_ε^2 .

3.2

Looking at the two correlograms:

- First AR(2) : The ACF oscillates above and below zero before decaying.

This is a signature of complex conjugate roots. Complex roots cause the autocorrelation to decay geometrically with a sinusoidal component, producing oscillations.

- Second AR(2) : The ACF decays smoothly to zero without oscillating.

This indicates real roots, because real roots produce a monotone geometric decay of the autocorrelation.

3.3

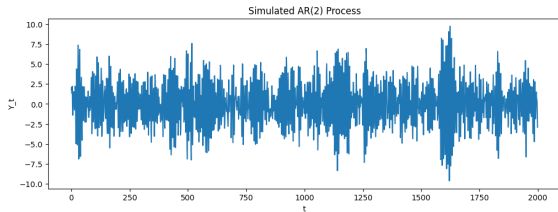
$$S(f) = \sigma_\varepsilon^2 \left| 1 - \phi_1 e^{-2\pi i f} - \phi_2 e^{-4\pi i f} \right|^2.$$

3.3

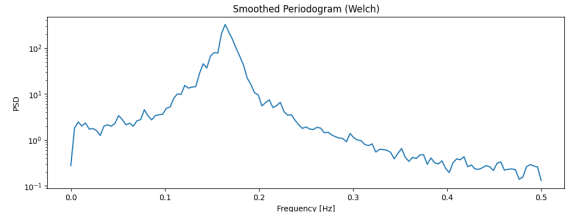
Since the AR(2) has complex roots, the process exhibits oscillatory behavior. The periodogram shows a peak.

3.4

$$\phi_1 = \frac{2 \cos \theta}{r}, \quad \phi_2 = -\frac{1}{r^2}.$$



Signal



Periodogram

Figure 2: AR(2) process

4 Sparse coding

The modulated discrete cosine transform (MDCT) is a signal transformation often used in sound processing applications (for instance, to encode an MP3 file). A MDCT atom $\phi_{L,k}$ is defined for a length $2L$ and a frequency localisation k ($k = 0, \dots, L - 1$) by

$$\forall u = 0, \dots, 2L - 1, \quad \phi_{L,k}[u] = w_L[u] \sqrt{\frac{2}{L}} \cos\left[\frac{\pi}{L} \left(u + \frac{L+1}{2}\right) \left(k + \frac{1}{2}\right)\right] \quad (3)$$

where w_L is a modulating window given by

$$w_L[u] = \sin\left[\frac{\pi}{2L} \left(u + \frac{1}{2}\right)\right]. \quad (4)$$

Question 4 *Sparse coding with OMP*

For the signal provided in the notebook, learn a sparse representation with MDCT atoms. The dictionary is defined as the concatenation of all shifted MDCT atoms for scales L in $[32, 64, 128, 256, 512, 1024]$.

- For the sparse coding, implement the Orthogonal Matching Pursuit (OMP). (Use convolutions to compute the correlation coefficients.)
- Display the norm of the successive residuals and the reconstructed signal with 10 atoms.

Answer 4

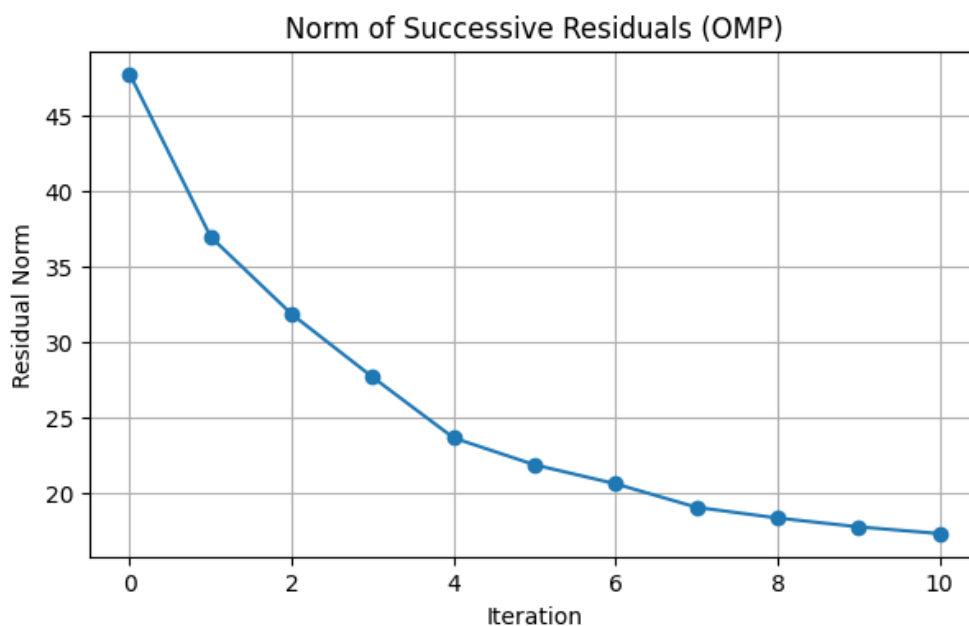


Figure 3: Norms of the successive residuals

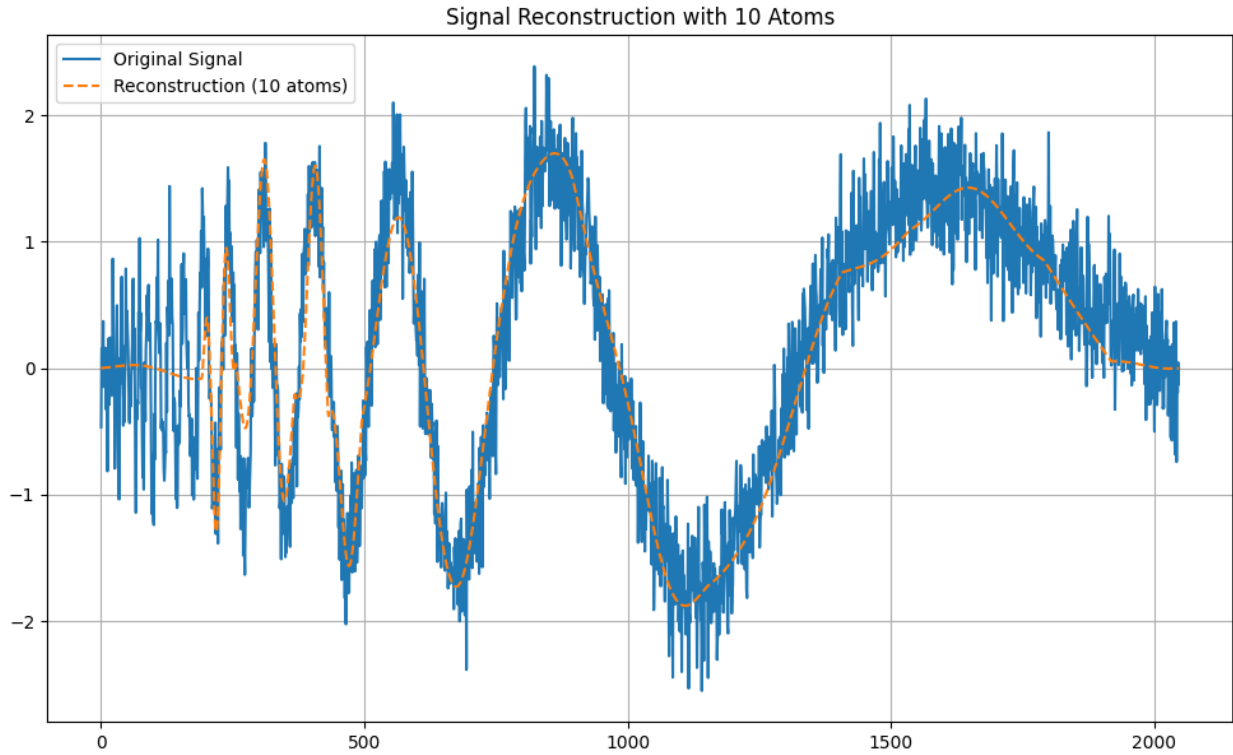


Figure 4: Reconstruction with 10 atoms

The 10-atom MDCT sparse representation successfully extracted the dominant, low-frequency structure of the original signal, removing the noise present in the signal.

However, the chosen dictionary and the constraint of only 10 atoms were insufficient to accurately represent the complex, high-frequency, time-localized content present in the initial segment of the signal.